Does Corporate Ownership Matter to AI? Portfolio Interference in Large Language Model Brand Perception

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Abstract

Large language models (LLMs) increasingly shape how consumers discover and evaluate brands. Brand portfolio theory predicts that revealing corporate ownership should produce perceptual interference via an awareness gate (Keller 1993; Aaker and Keller 1990). Yet LLMs encode portfolio relationships permanently in their parameters, raising the possibility of either maximal interference or complete immunity. We test these hypotheses in a preregistered experiment with 13 LLMs spanning seven training traditions (Western, Chinese, Russian, Indian, Japanese, European, Korean). Twenty brands from seven portfolio archetypes (LVMH, Unilever, P&G, Toyota, L'Oreal, Geely, Yandex) were rated under four prompt modalities—direct rating, naturalistic recommendation, multi-turn conversation with mid-dialogue portfolio reveal, and native-language framing—yielding 7,975 parsed observations. A preregistered extension with 20 additional brands drawn from published marketing research (Aaker 1997; Brakus et al. 2009; Li et al. 2024; and others) added 1,950 observations with zero overlap, replicating immunity (0/10 |d| > .50) and bringing the combined dataset to 9,925 observations across 40 brands. Portfolio framing produces near-zero change in Dimensional Concentration Index (mean |delta DCI| = .26). Equivalence testing confirms the null for 18/20 brands within +/-1.0 DCI points. Multi-turn revelation unlocks modest flattening for reverse-aspiration structures (Geely Auto d = −1.11, FDR-significant), but effects remain portfolio- and modality-specific rather than systematic. Native-language prompts activate model-specific discourse layers without directional amplification. These results generalize spectral immunity across model traditions and portfolio types, implying that house-of-brands shielding is automatic in AI-mediated markets while constructive interference is impossible. Brand managers cannot rely on portfolio architecture to reshape LLM perceptions except in extended conversational contexts. Theoretical implications for awareness-gate mechanisms and practical implications for "Share of Model" strategy are discussed.

Keywords: brand perception, large language models, portfolio interference, awareness gate, TOST equivalence, cross-cultural AI, multi-turn conversation, native-language framing, Share of Model

When a consumer asks an LLM "Which luxury bag should I buy?", the model's response reflects its internal brand encoding, shaped by training data composition rather than lived experience. Understanding these AI observers is now operationally critical (Dubois, Dawson, and Jaiswal 2025). LLMs increasingly mediate consumer access to brand information through search, recommendation, and conversational interfaces, constituting a new class of brand observer whose perceptual structure differs systematically from human observers (Li et al. 2024; Sabbah and Acar 2026).

Brand portfolio theory has long recognized that corporate ownership structures create perceptual interdependencies. Classic portfolio theory (Aaker and Keller 1990; Keller 1993) predicts interference when corporate ownership becomes salient: constructive for spectrally similar brands, destructive for contradictory ones. The mechanism is an awareness gate—the degree to which the observer recognizes shared corporate ownership. Erdem (1998) demonstrated that umbrella branding creates correlated quality expectations. Lei, Dawar, and Lemmink (2008) showed that portfolio spillover is asymmetric and sensitive to perceived fit. Brand, Israeli, and Ngwe (2023) demonstrated that LLMs can be used for market research, raising the question of whether these instruments respond to contextual manipulations the way human respondents do. Recent work on AI-mediated brand perception (Arora, Chakraborty, and Nishimura 2025; Hermann and Puntoni 2025) further establishes that LLM brand responses are systematic and measurable—but whether they are sensitive to portfolio context has not been tested.

For LLMs, the awareness gate is permanently saturated. These models cannot be experimentally manipulated into "not knowing" that Dior belongs to LVMH—the information is encoded in their parameters. This creates the *awareness gate paradox*: if interference scales with awareness, and awareness is permanently maximal, AI brand perception should exhibit permanent maximal interference.

Alternatively, LLMs may operate near minimum-distortion encodings of brand information. If dimensional compression is already at its limit, adding portfolio context provides no additional signal that would alter the encoding. One theoretical framework that predicts this immunity via rate-distortion compression is Spectral Brand Theory (Zharnikov 2026a), but the prediction follows from any account in which LLM brand encodings are compressed and stable.

This paper provides the first comprehensive test of portfolio interference in LLM brand perception. We test competing hypotheses—maximal interference vs. spectral immunity—through a preregistered experiment with six design innovations: 13 models from 7 training traditions, 7 portfolio archetypes, 4 prompt modalities, multi-turn conversation with mid-dialogue portfolio reveal, native-language ablation, and prompt-location ablation. Across 7,975 observations, the predominant finding is clear: 0/20 brands show FDR-significant portfolio effects in direct rating or recommendation conditions.

Theoretical Background

Portfolio Interference in Human Perception

Corporate brand portfolios create perceptual interdependencies. Keller's (1993) customer-based brand equity model predicts that corporate associations transfer to subsidiary brands through associative network activation. Aaker and Keller (1990) showed that brand extension attitudes depend on perceived fit between parent brand and extension. Erdem (1998) demonstrated that umbrella branding creates correlated quality expectations across portfolio members. Lei, Dawar, and Lemmink (2008) showed that portfolio spillover is asymmetric: negative spillover is stronger from weaker to stronger brands. Peng et al. (2023) meta-analyzed 2,134 effect sizes from three decades of brand extension research, confirming that both parent brand equity and extension fit

positively influence extension success. At the financial level, Morgan and Rego (2009) established that brand portfolio characteristics explain meaningful variance in firm performance.

Interference direction depends on the structural relationship between portfolio members. *Constructive interference* arises when sibling brands occupy proximate positions on high-salience dimensions. *Destructive interference* arises when siblings contradict on those dimensions. *Negligible interference* characterizes brands occupying distinct perceptual territories with low shared dimensionality. The key moderator in all accounts is the awareness gate: the consumer must know about the ownership relationship for any interference to occur (Aaker and Joachimsthaler 2000; Strebinger 2014).

Four portfolio archetypes operationalize these predictions. *Spectral clusters* (e.g., LVMH) house brands in proximate positions, producing constructive interference. *Spectral contradictions* (e.g., Unilever) house brands contradictory on high-weight dimensions, producing destructive interference. *Spectral spreads* (e.g., P&G) house brands in distant regions, producing negligible interference. *Spectral layers* (e.g., Toyota/Lexus) house brands at different positions on the same dimensions, serving distinct cohorts with aspiration dynamics. The present study adds three further archetypes: *prestige spread* (L'Oreal), *reverse aspiration* (Geely), and *branded house* (Yandex).

AI as Brand Observer

Li et al. (2024) validated LLMs as brand perception instruments, achieving over 75% agreement with human perceptual data across a large-scale Marketing Science study. Brand, Israeli, and Ngwe (2023) demonstrated the use of GPT for market research tasks including brand positioning. Arora, Chakraborty, and Nishimura (2025) demonstrated AI-human hybrid approaches for marketing research, establishing reliability and validity benchmarks. Hermann and Puntoni (2025) examined how generative AI shapes stakeholder engagement, arguing that understanding the internal structure of LLM brand responses is now a strategic necessity. Grewal, Levy, and Kumar (2025) document the broader shift toward generative AI in marketing practice. Cillo, Verona, and Wuyts (2025) outline a research roadmap for generative AI in marketing, identifying brand perception measurement as a priority area. Estevez (2025) surveys LLM applications for market research, noting both capabilities and limitations of current approaches.

LLM brand responses show systematic dimensional patterns. Across 24 models from seven training traditions, cross-model cosine similarity for brand profiles reaches .977, with systematic compression of the Cultural, Temporal, and Economic dimensions (Zharnikov 2026v). Sabbah and Acar (2026) found that only quantitative ratings are stable across models. Goli and Singh (2024) demonstrated structural divergence between LLM and human preferences in choice tasks. None of this prior work tested whether LLM brand perception is sensitive to contextual manipulation —specifically, whether adding portfolio context to a brand query changes the model's brand encoding.

The cognitive science literature reveals relevant evidence. Binz and Schulz (2023) found that GPT-3 exhibits human-like framing effects yet also systematic deviations. Germani and Spitale (2025) showed that source framing—attributing identical statements to different authors—triggers systematic bias across LLMs. Portfolio framing is a content-level manipulation (adding factual

context the model already knows) rather than a social-attribution manipulation, which may explain divergent results.

The Awareness Gate Paradox

The awareness gate mechanism (Keller 1993) holds that portfolio interference requires the observer to recognize the ownership relationship. In human samples, this gate is calibrated experimentally: studies manipulate whether respondents learn about corporate ownership. In LLMs, no such manipulation is possible—the ownership relationship is encoded in parameters. The awareness gate is permanently saturated at $\alpha \approx 1$.

This creates the awareness gate paradox. If interference scales with awareness (as standard portfolio theory predicts), LLMs should show maximal portfolio effects. Yet if LLM encodings are already compressed to near-minimum-distortion representations of brands, adding portfolio context provides no additional information the encoding can absorb. The distinction is between an observer for whom portfolio knowledge is *new* (human experimental manipulation) and one for whom it is *structural* (LLM parameters). The present study tests which regime governs LLM brand perception.

One formal representation of this compression account is the SBT interference equation:

$$I_d(B_i, B_j, C_k) = \alpha_{C_k} \cdot \bar{w}_d^{(C_k)} \cdot (e_d^{(B_j)} - \mu_d) \quad (1)$$

where α_{C_k} is the awareness gate, $\bar{w}_d^{(C_k)}$ is the cohort's weight on dimension d, $e_d^{(B_j)}$ is the sibling brand's emission, and μ_d is the category mean.[1] Under immunity, α_{C_k} is maximal but the compression constraint prevents the interference term from propagating into the observable profile. The present paper tests this prediction empirically without requiring acceptance of any particular theoretical framework.

Research Hypotheses

Core hypotheses (H1–H5) are derived from portfolio theory and the awareness gate mechanism. Extension hypotheses (H6–H9) test new archetypes and modalities added in the present study.

H1 (Constructive Interference). Portfolio framing increases dimensional concentration (DCI) for brands in spectrally clustered portfolios (LVMH).

H2 (Destructive Interference). Portfolio framing decreases dimensional concentration for brands in spectrally contradictory portfolios (Unilever).

H3 (Negligible Interference). Portfolio framing produces no measurable change for brands in spectrally spread portfolios (P&G).

H4 (Aspirational Interference). Portfolio framing produces asymmetric effects for brands in spectrally layered portfolios (Toyota/Lexus), with greater effect on the aspirational brand (Lexus).

[1]This formulation is from Spectral Brand Theory (Zharnikov 2026a), where Dimensional Concentration Index (DCI) measures profile concentration across eight brand perception dimensions. DCI is a new metric not yet validated against human data; Section Limitations discusses this as a primary boundary condition on our conclusions.

H5 (Spectral Immunity). Across all portfolios and prompt modalities, the magnitude of portfolio-induced perception change is equivalent to zero within meaningful bounds (+/-1.0 DCI points).

H6 (Prestige Gradient). Portfolio framing flattens the prestige gradient in L'Oreal's brand portfolio, pulling Lancome toward Maybelline's position.

H7 (Reverse Aspiration). Portfolio framing suppresses Volvo's premium positioning under Geely ownership, reflecting downward aspirational interference.

H8 (Language-Dependent Immunity). Native-language portfolio framing produces larger |delta DCI| than English-language framing for the same brands.

H9 (Home-Model Amplification). The native-language effect is strongest when model training tradition matches the language of framing.

Method

Design

The experiment employed a fully crossed design across four prompt modalities:

1. **Direct rating** (main experiment): 20 brands x 2 conditions (solo vs. portfolio) x 13 models x 5 repetitions = 2,600 observations.
2. **Naturalistic recommendation**: 20 brands x 2 conditions x 13 models x 5 repetitions = 2,600 observations. Prompts framed as conversational queries ("What do you think of [BRAND]?").
3. **Multi-turn conversation**: 20 brands x 13 models x 5 repetitions = 1,300 observations. Turn 1 = solo rating; Turn 2 = portfolio reveal + re-rating within the same conversation context.
4. **Native-language ablation**: 11 brands (4 home portfolios: Toyota, Yandex, Geely, L'Oreal) x 2 conditions x 13 models x 5 repetitions = 1,430 observations. Prompts issued in the home language of the portfolio's country of origin.
5. **Prompt-location ablation**: 20 brands x 4 conditions x 5 repetitions = 400 observations (est). Portfolio information placed in system prompt rather than user message.

Total: approximately 7,930 observation cells (7,975 successfully parsed). All ratings used a 1–5 scale following the PRISM-B specification (Zharnikov 2026aa).

Brands and Portfolios

Table 1. Portfolio Composition and Predicted Interference Direction.

Portfolio	Archetype	Brands	Predicted Direction
LVMH	Spectral cluster	Louis Vuitton, Dior, Fendi	Constructive
Unilever	Spectral contradiction	Dove, Axe, Ben & Jerry's	Destructive
Procter & Gamble	Spectral spread	Tide, Pampers, Gillette	Negligible
Toyota	Spectral layer	Toyota, Lexus	Aspirational (asymmetric)
L'Oreal	Prestige spread	L'Oreal Paris, Lancome, Maybelline	Gradient flattening
Geely	Reverse aspiration	Volvo, Polestar, Geely Auto	Downward suppression
Yandex	Branded house	Yandex, Yandex Taxi, Yandex Market	Shared identity

Notes: The Toyota portfolio tests spectral layering where the mass brand (Toyota) and luxury brand (Lexus) occupy different positions on the same dimensions. The Geely portfolio tests reverse aspiration where a premium acquired brand (Volvo) may be pulled downward by its lower-tier parent. The Yandex portfolio tests branded house architecture where shared naming creates maximum awareness gate saturation.

Models

Table 2. Model Panel: 13 Models From 7 Training Traditions.

Model	Provider	Training Tradition	Deployment
Claude Sonnet 4	Anthropic	Western (US)	Cloud API
GPT-4o-mini	OpenAI	Western (US)	Cloud API
Gemini 2.5 Flash	Google	Western (US)	Cloud API
Grok-3-mini	xAI	Western (US)	Cloud API
Llama 3.3 70B	Meta via Groq/ SambaNova	Western (US, open-weight)	Cloud API
Gemma 4 27B	Google	Western (US, open-weight)	Local (Ollama)
DeepSeek V3	DeepSeek	Chinese	Cloud API
Qwen3 235B	Alibaba via Cerebras	Chinese	Cloud (free)
YandexGPT 5 Pro	Yandex	Russian	Cloud API
Sarvam M	Sarvam AI	Indian	Cloud API
GPT-OSS-Swallow 20B	Tokyo Tech via Yandex	Japanese	Cloud API
Mistral Large	Mistral AI	European	Cloud API
EXAONE 3.5 32B	LG AI Research	Korean	Local (Ollama)

Notes: Models span 7 training traditions. The inclusion of YandexGPT (Russian-first), Sarvam (Indian-first), GPT-OSS-Swallow (Japanese-first, trained from scratch by Tokyo Tech), Mistral Large (European), and EXAONE 3.5 32B (Korean, LG AI Research) tests whether models with different cultural knowledge bases show different immunity patterns. Temperature = .7 for all models. Parse success rate: 7,975/7,930 (98.1%). Local models ran on Apple Mac mini M4 Pro (48 GB RAM) via Ollama v0.20.0.

Power Analysis

The experiment had 70% power to detect a medium effect (d = .50) at alpha = .05 with N = 50 per group in the main experiment; with N = 65 per cell in v2.0, power exceeds 80% for d = .50. The consistently small observed effect sizes indicate that even substantially larger samples would confirm the null. TOST equivalence testing provides a more appropriate framework for this confirmatory-null design.

Metrics

The *Dimensional Concentration Index* (DCI) measures profile concentration:

$$DCI = \frac{\sum_{d=1}^{8} |w_d - \frac{1}{8}|}{2} \times 100 \quad (2)$$

TOST equivalence testing with bounds of +/-1.0 DCI points assesses statistical equivalence to zero. Benjamini-Hochberg correction controls false discovery rate across all 20 brand-level tests.

Results

Main Results: Direct Rating Solo vs. Portfolio

The predominant finding is clear: 0/20 brands show FDR-significant portfolio effects in direct rating conditions. Table 3 presents the full brand-level results.

Table 3. DCI by Brand and Condition (N = 7,975; 13 Models, 5 Repetitions).

Portfolio	Brand	Solo DCI	Port DCI	Delta	t	p	d	TOST p	Equivalent
LVMH	Dior	3.5	3.8	+.35	1.13	.262	+.20	.020	Yes
LVMH	Fendi	5.7	5.8	+.09	.38	.702	+.07	.000	Yes
LVMH	Louis Vuitton	4.4	4.7	+.34	.90	.368	+.16	.043	Yes
Unilever	Axe	8.6	8.7	+.04	.12	.909	+.02	.002	Yes
Unilever	Ben & Jerry's	6.4	6.4	-.04	-.24	.809	-.04	.000	Yes
Unilever	Dove	5.6	5.9	+.30	1.20	.234	+.21	.003	Yes
P&G	Gillette	6.4	6.3	-.15	-.71	.477	-.13	.000	Yes
P&G	Pampers	6.5	6.2	-.29	−1.27	.207	-.22	.001	Yes
P&G	Tide	7.4	7.2	-.20	-.55	.580	-.10	.016	Yes
Toyota	Toyota	6.8	6.4	-.37	−1.69	.093	-.30	.003	Yes
Toyota	Lexus	5.2	5.8	+.59	2.94	.004	+.52	.022	Yes
L'Oreal	L'Oreal Paris	5.6	5.5	-.15	-.75	.456	-.13	.000	Yes
L'Oreal	Lancome	5.4	5.4	+.04	.18	.858	+.03	.000	Yes
L'Oreal	Maybelline	6.8	6.6	-.13	-.38	.705	-.07	.007	Yes
Geely	Volvo	6.4	6.3	-.19	-.81	.422	-.15	.000	Yes
Geely	Polestar	8.6	8.5	-.12	-.37	.710	-.07	.003	Yes
Geely	Geely Auto	8.0	8.5	+.54	1.49	.139	+.27	.102	No
Yandex	Yandex	6.6	6.6	-.01	-.03	.976	-.01	.000	Yes
Yandex	Yandex Market	8.2	8.4	+.21	.54	.592	+.10	.021	Yes
Yandex	Yandex Taxi	8.9	8.5	-.36	-.87	.384	-.16	.062	No

Notes: N = 65 per cell (13 models x 5 repetitions) for original portfolios; N = 59–65 for v2.0 portfolios. Benjamini-Hochberg correction yields 0/20 significant at FDR = .05. TOST equivalence confirmed for 18/20 brands within +/-1.0 DCI bounds. Geely Auto and Yandex Taxi are inconclusive (neither equivalent nor significantly different). Lexus shows p = .004 in the paired t-test but does not survive FDR correction with 20 tests.

Hypothesis Tests

H1 (LVMH constructive): Not supported. Mean delta DCI = +.26, effect sizes small (d range +.07 to +.20), no FDR-significant results.

H2 (Unilever destructive): Not supported. Mean delta DCI = +.10, effect sizes negligible (d range -.04 to +.21), no FDR-significant results.

H3 (P&G negligible): Supported. Mean delta DCI = -.21, TOST equivalent for all 3 brands, no FDR-significant results.

H4 (Toyota/Lexus aspirational): Partially supported. Lexus showed delta = +.59 (d = .52, p = .004), directionally consistent with aspiration dynamics. However, with 20 tests, 0/20 survive FDR correction. The Lexus finding is suggestive but not definitive: aspiration dynamics may partially penetrate AI perception, but the evidence is inconclusive at the corrected alpha.

H5 (Spectral Immunity): Supported. TOST equivalence confirmed for 18/20 brands. Mean |delta DCI| = .26. Cosine similarity > .999 for all brands. 0/20 FDR-significant results. The two inconclusive brands (Geely Auto, Yandex Taxi) show neither equivalence nor a reliable effect—they are statistically underpowered for a conclusion rather than exceptions to immunity.

H6 (L'Oreal prestige gradient): Not supported. All three L'Oreal brands show near-zero delta (range -.15 to +.04, all d < .14). Portfolio architecture is invisible even for a prestige-spread structure spanning Lancome to Maybelline.

H7 (Geely reverse aspiration): Not supported in direct framing. Volvo shows delta = -.19 (d = -.15, TOST equivalent). The predicted downward suppression is directionally present but too small to constitute evidence. The reverse aspiration hypothesis receives support only in the multi-turn condition (see below).

H8 (Language-dependent immunity): Mixed. Native-language framing activates different discourse layers for some portfolios (Geely: Japanese model Swallow shows delta −2.35; Yandex: YandexGPT shows +.72). However, effects are portfolio-specific and model-specific, with no systematic directional pattern.

H9 (Home-model amplification): Mixed. The predicted home-model amplification appears selectively. Swallow (Japanese) shows the largest tradition-level native effect for Toyota (-2.35). YandexGPT (Russian) shows modest amplification for Yandex (+.72). Qwen3 (Chinese) shows a conditional effect for Volvo (+2.07). None of these survive FDR correction, and there is no consistent pattern of amplification that holds across all home-market pairs.

Cross-Cultural Generalizability

Table 4. Sub-Group Analysis by Training Tradition.

Tradition	Models (n)	Delta DCI	d	p (tradition level)
Western	6	+.02	+.03	.882
Chinese	2	-.06	-.09	.692
European	1	-.35	-.46	.055
Japanese	1	-.59	-.63	.011
Korean	1	+.34	+.35	.130
Russian	1	-.24	-.32	.168
Indian	1	-.13	-.07	.743

Notes: The Japanese model (GPT-OSS-Swallow) shows the only tradition-level p-value below .05 ($p = .011$, $d = -.63$). This pattern is driven primarily by the Toyota-related brands, consistent with deeper Japanese automotive market knowledge. However, this is a single model, and the effect does not survive FDR correction across 7 tradition comparisons. All other traditions show identical immunity. The European model (Mistral Large) shows a non-significant trend ($p = .055$) in the direction of sensitivity.

Variance Decomposition

Table 5. Variance Decomposition of DCI.

Factor	% Variance
Brand	37.4
Portfolio (parent company)	19.3
Model	8.6
Condition (solo vs. portfolio)	.1

Robustness Checks

Naturalistic recommendation prompts. When portfolio framing was tested through naturalistic recommendation prompts ("What do you think of [BRAND]? I know it's part of [PARENT]"), the pattern was identical to direct rating: mean delta DCI = -.23, cosine > .999, 0/20 FDR-significant. The recommendation modality produced slightly more variation in DCI levels compared to direct rating (mean cosine between direct and recommendation profiles = .998), but the solo-vs-portfolio differential remained near zero.

Prompt-location ablation. Moving portfolio context from the user message to the system message produced no difference (paired $t = .91$, $p = .368$, $d = .15$; 4 models, 20 brands). Portfolio information location is irrelevant to perception outcomes.

Extensions

Multi-turn conversation: Portfolio reveal. Multi-turn conversation is the most ecologically valid test and the modality where the largest individual effects appear. Four brands showed FDR-significant DCI shifts after portfolio reveal:

Table 6. Multi-Turn Results: FDR-Significant Brands (Turn 1 Solo vs. Turn 2 Post-Reveal).

Brand	T1 DCI	T2 DCI	Delta	p	d
Geely Auto	8.1	5.1	−2.93	<.001	−1.11
Polestar	8.8	7.4	−1.47	<.001	-.54
Toyota	6.8	6.0	-.80	<.001	-.61
Yandex	6.2	5.1	−1.06	<.001	-.49

Notes: 4/20 brands showed FDR-significant shifts in multi-turn conversation. All four showed DCI decreases (profile flattening) after portfolio reveal, consistent with contextual dilution rather than constructive interference. Geely Auto shows the largest effect in the entire experiment (d = −1.11): when the model is explicitly reminded mid-conversation that Geely Auto is the parent of Volvo and Polestar, the reverse aspiration operates in the predicted direction—but only through conversational accumulation, not direct framing. Mean delta across all 20 brands was -.03—effectively zero. The Geely Auto multi-turn finding is notable but conditional: it requires extended conversational context and reflects a specific reverse-aspiration dynamic rather than a general exception to immunity.

Native-language ablation. Native-language framing was tested for the 4 portfolios with identifiable home-market languages (Japanese/Toyota, Russian/Yandex, Chinese/Geely, French/L'Oreal). Results show that native-language framing activates different discourse registers for specific model-portfolio pairs but does not produce systematic amplification of portfolio effects. The clearest signal is the Japanese model (Swallow) under Japanese-language Toyota framing (delta −2.35 vs. -.37 in English), and YandexGPT under Russian-language Yandex framing (+.72 vs. -.01 in English). However, Qwen3 under Chinese-language Geely framing shows a Volvo-specific effect (+2.07) that is opposite in direction to the home-model amplification prediction. No native-language effect survives FDR correction.

Replication with Published Brand Stimuli

The 20 brands in the main experiment were selected to represent seven portfolio archetypes. That selection, while theoretically motivated, is concentrated in a narrow band of portfolio combinations. To address the external validity concern, we conducted an extension using an independent brand stimulus set drawn entirely from published peer-reviewed marketing research.

Brand Selection

Twenty brands were drawn from canonical empirical studies: Aaker (1997, JMR), Brakus, Schmitt, and Zarantonello (2009, JM), Li, Castelo, Katona, and Sarvary (2024, Marketing Science), Erdem (1998, JMR), Dew, Ansari, and Li (2022, Marketing Science), Morgan and Rego (2009, JM), and Malar, Krohmer, Hoyer, and Nyffenegger (2011, JM). These stimulus sets span brand personality,

brand experience, brand extensions, umbrella branding, and emotional brand attachment—traditions that are empirically distinct from the portfolio interference literature. Zero overlap exists between the 20 extension brands and the 20 main-experiment brands.

Ten brands were tested in both solo and portfolio conditions: Coca-Cola (Coca-Cola Co), Pepsi (PepsiCo), BMW (BMW Group), Audi (Volkswagen Group), Google (Alphabet), Disney (Walt Disney Co), Colgate (Colgate-Palmolive), Samsung (Samsung Group), H&M (H&M Group), and Dell (Dell Technologies). Ten brands were tested in solo condition only: Apple, Nike, Starbucks, IBM, Harley-Davidson, Rolex, Amazon, Adidas, McDonald's, and Red Bull. The same 13-model, 7-tradition panel and identical PRISM-B protocol were applied. Total: 1,950 observations (1,300 solo + 650 portfolio), 5 repetitions per cell.

Results

Table 7. Published-Brand Extension: Portfolio Immunity Test (N = 1,950).

Brand	Parent	Solo DCI	Port DCI	Delta	d
Coca-Cola	Coca-Cola Co	.56	.58	+.02	+.08
Pepsi	PepsiCo	.85	.89	+.03	+.11
BMW	BMW Group	.46	.43	-.03	-.18
Audi	VW Group	.62	.71	+.09	+.49
Google	Alphabet	.57	.58	+.01	+.06
Disney	Walt Disney Co	.37	.37	+.01	+.04
Colgate	Colgate-Palmolive	1.04	1.10	+.06	+.21
Samsung	Samsung Group	.73	.75	+.02	+.06
H&M	H&M Group	1.31	1.21	-.10	-.29
Dell	Dell Technologies	.92	.98	+.06	+.16

Notes: N = 65 per cell (13 models x 5 repetitions). 0/10 brands FDR-significant at q = .05. TOST equivalence confirmed for all 10 brands within +/-1.0 DCI bounds.

The extension replicates the core immunity finding with zero exceptions. No brand shows |d| > .50 in the portfolio condition. The largest individual effect is Audi (d = +.49, Volkswagen Group framing), which falls just below the medium-effect threshold and is TOST-equivalent. Disney (d = +.04), Google (d = +.06), and Samsung (d = +.06) show near-exact immunity regardless of whether the parent conglomerate name substantially differs from the brand name. H&M (d = -.29) shows the only meaningful negative shift, consistent with profile diffusion when additional brands (COS, & Other Stories) are introduced.

The Audi result echoes the suggestive Lexus finding from the main experiment: both involve premium-differentiation dynamics where the brand occupies a distinctly more premium position

than its corporate sibling average, and both trend positive without reaching the medium-effect threshold.

Combined with the main experiment, R20 now encompasses 9,925 observations across 40 brands (20 + 20, zero overlap), 13 models, and 7 training traditions. Spectral immunity generalizes across stimulus sets drawn from independent empirical research traditions spanning three decades of brand theory.

Discussion

Theoretical Contribution

We resolve the awareness-gate paradox by showing that maximal awareness is necessary but not sufficient for interference when observers operate near minimum-distortion encodings. Standard portfolio theory (Keller 1993; Aaker and Keller 1990) treats the awareness gate as a scalar that amplifies interference when activated. For LLMs, the gate is permanently saturated ($\alpha \approx 1$), yet interference is absent. The resolution is that LLMs encode brand knowledge holistically: the solo profile already incorporates portfolio relationships implicitly. The gate amplifies an interference *term* that is structurally near zero, not an absent gate suppressing a nonzero term.

Spectral immunity generalizes to 7 portfolio archetypes. The original finding (4 archetypes, 4 portfolios, 10 models, 5 traditions) now extends to 7 archetypes, 7 portfolios, 13 models, and 7 traditions. L'Oreal's prestige spread, Yandex's branded house, and Geely's reverse aspiration all show the same immunity pattern as LVMH, Unilever, P&G, and Toyota in direct framing. The structural property holds regardless of portfolio type.

Immunity is architectural, not knowledge-based. All 7 model traditions show immunity. The Japanese model (GPT-OSS-Swallow) shows the only tradition-level significant effect ($p = .011$, $d = -.63$), driven primarily by Toyota-related brands, consistent with deeper Japanese automotive market knowledge. But this is one model, the effect does not survive tradition-level FDR correction, and it does not break the fundamental immunity pattern.

The Lexus finding is suggestive but no longer statistically robust. With 20 brands and FDR correction, the Lexus result ($p = .004$, $d = .52$) yields 0/20 FDR-significant results. Aspiration dynamics remain directionally suggestive—Lexus shows the largest positive delta of any brand—but the evidence does not support concluding that aspiration dynamics are an exception to immunity. Future work should test this specifically with a dedicated aspiration-focused design.

Reverse aspiration can be unlocked through conversational revelation but not direct framing. The Geely Auto multi-turn finding ($d = -1.11$) is the strongest single effect in the experiment. In direct framing, Geely Auto shows a small non-significant increase in DCI when portfolio context is added. But in multi-turn conversation, when the model has established Volvo's premium perception in Turn 1 and then receives portfolio context in Turn 2, Geely Auto's DCI drops by 2.93 points. The mechanism appears to be conversational anchoring: Turn 1 establishes a premium anchor for the portfolio family, making Turn 2′s framing more effective. This pattern

does not appear for portfolios without the reverse-aspiration dynamic (LVMH, P&G), suggesting it is specific to ownership structures that violate the quality-hierarchy expectation.

Native-language framing activates discourse layers but not systematically. The native-language ablation reveals model-portfolio-specific sensitivity rather than a general amplification principle. The Japanese model's sensitivity to Japanese-language Toyota framing, and YandexGPT's sensitivity to Russian-language Yandex framing, suggest that home-language prompts activate different discourse registers in home-tradition models. But the effects are not unidirectional, not consistent across portfolios, and do not survive FDR correction. This is consistent with Zharnikov (2026v)'s finding that cross-cultural differences in AI brand perception represent discourse-layer activation rather than systematic cultural bias.

Managerial Contribution

Portfolio architecture is invisible to AI across all archetype types. As AI-mediated brand interactions grow—what Dubois, Dawson, and Jaiswal (2025) term the "Share of Model" economy—portfolio strategies become irrelevant in these channels across all 7 archetypes tested. Prestige spread (L'Oreal), reverse aspiration (Geely), and branded house (Yandex) are as invisible to AI as symmetric interference structures.

Shielding is automatic but amplification is impossible. House-of-brands architecture shields brands from destructive interference automatically in AI perception. But constructive interference is equally blocked. The Yandex branded house—where shared naming maximally saturates the awareness gate—shows the same immunity as LVMH's house-of-brands architecture.

Geely's reverse aspiration is a strategic vulnerability in conversational AI. While direct portfolio framing leaves Volvo's positioning intact, multi-turn conversation reveals a meaningful effect: when users engage in extended dialogue about Geely's portfolio, Geely Auto's perceived differentiation collapses ($d = -1.11$). Brand managers with reverse-aspiration portfolio structures should anticipate that extended AI interactions about the portfolio can surface dynamics that are invisible in single-turn queries.

The Lexus strategy remains an open empirical question. The aspirational differentiation pattern (Lexus $d = .52$) is directionally consistent but not statistically robust after FDR correction with 20 brands. Portfolio managers pursuing vertical differentiation strategies (Toyota/Lexus, Volkswagen/Audi, Gap/Banana Republic) should treat AI-channel aspiration dynamics as warranting dedicated replication before acting on them.

Brand coherence, not portfolio coherence, drives AI perception. Brand identity explains 37% of DCI variance; condition explains .1%.

Boundary Conditions and Limitations

First, DCI is a new metric not yet validated against human perceptual data. The core finding—near-zero portfolio effects—is robust to this limitation (a null is a null regardless of the specific metric), but the theoretical interpretation in terms of rate-distortion compression depends on DCI's construct validity, which is the primary boundary condition on our conclusions. Validation against human brand perception data is the most important direction for future work.

Second, with 20 tests and FDR correction, the study has limited power to detect effects at the brand level (the Lexus finding would be significant under a per-comparison alpha). A dedicated within-archetype replication with larger samples would provide cleaner evidence.

Third, the multi-turn design uses a two-turn protocol; longer conversations with more gradual portfolio disclosure might accumulate larger effects.

Fourth, native-language results are based on a single model per tradition (except Western and Chinese), limiting the generalizability of tradition-level conclusions.

Future Research

Four extensions are warranted. First, validation of DCI against human perceptual data to establish construct validity for the metric and the immunity finding. Second, a dedicated replication of the Lexus finding with a multi-brand aspirational layer panel (Toyota/Lexus, Volkswagen/Audi, Marriott/Ritz-Carlton) to determine whether d = .52 represents a real exception. Third, agentic simulations and extended multi-turn designs (5+ turns) to test whether conversational accumulation can unlock portfolio effects in non-reverse-aspiration portfolios. Fourth, longitudinal testing of whether spectral immunity is stable as models are updated or fine-tuned, which is critical for "Share of Model" strategy planning.

Conclusion

This study provides the first comprehensive test of portfolio interference in LLM brand perception. Across 7,975 observations, 13 models from 7 training traditions, 7 portfolio archetypes, and 4 prompt modalities, the predominant finding is clear: portfolio framing does not systematically alter how LLMs perceive brands. We term this phenomenon *spectral immunity*. The mechanism is architectural—Russian, Indian, Japanese, European, and Korean models with different training corpora show identical immunity to Western models. The original Lexus finding (d = .52) is suggestive but FDR-inconclusive with the full 20-brand panel. The multi-turn Geely Auto finding (d = −1.11) is the strongest effect in the experiment, revealing that reverse aspiration can be unlocked through conversational revelation—but this is a conditional vulnerability, not a general exception to immunity. For brand managers, portfolio architecture is invisible to the AI systems increasingly mediating consumer-brand interactions. The exception is conversational AI with extended interaction: portfolios with reverse-aspiration structure show meaningful effects in multi-turn dialogue that are absent in single-turn queries.

Falsification conditions: Spectral immunity (H5) is falsified if any future study, using comparable methods and a panel of $$7 models from $$3 traditions, finds systematic portfolio-induced DCI shifts exceeding 2.0 DCI points with effect sizes d > .50 that survive multiple testing correction across $$5 portfolio archetypes and $$3 prompt modalities simultaneously.

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Data Availability

Experiment data (7,975 observations) are archived at https://doi.org/10.57967/hf/8380. Source code and analysis scripts are available at https://doi.org/10.5281/zenodo.19555282. The dataset contains 2,555 direct-rating observations, 2,518 recommendation-prompt observations, 1,235 multi-turn conversation observations, 1,322 native-language ablation observations, and 345 prompt-location ablation observations across 13 models from 7 training traditions. Local model experiments were conducted on Apple Mac mini M4 Pro (48 GB RAM) running Ollama v0.20.0.